

Containerlab Network Troubleshooting Guide

Diagnosing & Resolving Path MTU Mismatches, TCP Stalls, and Jumbo Frame Drops

1. Executive Summary & Diagnostic Overview

In container-based emulation environments like **containerlab**, network engineers often encounter a deceptive issue: **ping** and **control protocol handshakes** work perfectly, but **bulk TCP transfers** (e.g., **iperf3**) stall completely at 0 bits/sec.

The Symptom Matrix		
Traffic Type	Observed Status	Reason
ping (ICMP)	PASSING	Default packet size is 64 bytes (fits within any link MTU).
TCP Handshake (SYN/ACK)	PASSING	Control frames are ~60 bytes; connection opens successfully.
UDP Throughput (iperf3 -u)	SENDER PASS / RCVR 0	UDP lacks flow control; sender transmits, but receiver drops silently if MTU/checksum fails.
TCP Throughput (iperf3 -c)	STALLS (0 bps)	Full MSS data packets (e.g., 9,460B) exceed bottleneck MTU (1,500B) and are dropped.

2. Deep Dive: Root Cause Analysis

[Host 1] -----> [Intermediate Router / Bridge] -----> [Host 2]

eth1 MTU: 9500 Transit Interface MTU: 1500 eth1 MTU: 9500

MSS Advertised: 9460 [Drops Packets > 1500 Bytes] MSS Advertised: 9460

1. TCP Handshake: Negotiates MSS = 9460 Bytes (based on local host interfaces)

2. Control Messages: Transmits ~100B JSON headers ---> SAYS CONNECTED --->

3. Data Stream: Transmits 9460B Jumbo Frames ---> DROPPED AT ROUTER (DF Set) ---> Silent Blackhole

A. MSS Negotiation vs. Path Bottlenecks

During the TCP 3-way handshake, endpoints exchange their local interface **MSS (Maximum Segment Size)**. If a host has an interface set to Jumbo Frames (e.g., `mtu 9500`), it advertises an MSS of **9,460 bytes** (MTU minus 40 bytes for standard IP/TCP headers).

When an intermediate router or bridge link only supports standard Ethernet (`mtu 1500`), it cannot forward these 9,460-byte packets. Because IP packets generated by TCP have the **Don't Fragment (DF)** flag set, the intermediate router drops the oversized frames.

B. Path MTU Discovery (PMTUD) Blackholing

In a standard network, the dropping router sends an `ICMP Type 3, Code 4 (Fragmentation Needed)` back to the sender. However, in lab container environments, firewall rules, missing ICMP routing back, or container namespace restrictions often block these messages. The sending host never learns to shrink its MSS, resulting in continuous retransmissions (`Retr`) and complete throughput collapse.

3. Step-by-Step Diagnostic Workflow

Step 1: Capture the TCP Handshake

Run `tcpdump` on the receiving host while initiating the `iperf3` test to inspect the negotiated MSS:

```
docker exec host2 tcpdump -vv -i eth1 port 8090
```

Sample Captured Output:

```
13:27:17 IP 10.0.100.2.35482 > 10.0.200.2.8090: Flags [S], cksum 0xa213, options [mss 9460,sackOK,...]
13:27:17 IP 10.0.200.2.8090 > 10.0.100.2.35482: Flags [S.], cksum 0x5dd0, options [mss
9460,sackOK,...]
```

Notice that both endpoints agreed to `mss 9460` because their local host interfaces were set to MTU 9500.

Step 2: Isolate the Path MTU using Ping with Don't Fragment (DF) Flag

Validate the exact maximum payload size that can traverse the end-to-end path without fragmentation:

```
# Test Standard 1500 MTU (1472 payload + 28 IP/ICMP headers = 1500 bytes)
docker exec host1 ping -c 2 -M do -s 1472 10.0.200.2

# Test Jumbo 9500 MTU (9472 payload + 28 IP/ICMP headers = 9500 bytes)
docker exec host1 ping -c 2 -M do -s 9472 10.0.200.2
```

Interpreting Ping Results

- If `-s 1472` succeeds but `-s 9472` fails, an intermediate link between host1 and host2 is capped at standard 1500 MTU.
- If `-s 1472` also fails, test smaller sizes (e.g., 1400, 1300) to find the tunnel/GRE overhead limit.

4. Solutions & Remediation

Option A: Set Uniform Host Interface MTU (Recommended for Standard Labs)

If intermediate network links cannot be changed, lower the MTU on the host container interfaces to 1500 so TCP naturally advertises `mss 1460` during handshake:

```
docker exec host1 ip link set dev eth1 mtu 1500
docker exec host2 ip link set dev eth1 mtu 1500
```

Option B: Enable TCP MSS Clamping on Router Containers

If hosts must retain jumbo MTU configuration but traverse 1500 MTU transit networks, enforce MSS clamping on intermediate Linux/FRR router nodes using `iptables`:

```
# Force router to rewrite SYN MSS options matching Path MTU
docker exec router1 iptables -A FORWARD -p tcp --tcp-flags SYN,RST SYN -j TCPMSS --clamp-mss-to-pmtu
```

Option C: Configure MTU Directly in Containerlab Topology File

Prevent manual post-spinup fixes by defining interface MTUs or running exec commands automatically inside your `clab.yaml` topology file:

```
name: net-lab
topology:
  nodes:
    host1:
      kind: linux
      image: alpine-networking
      exec:
        - ip link set dev eth1 mtu 1500
    router1:
      kind: linux
      image: frouting/frr:latest
      exec:
        - iptables -A FORWARD -p tcp --tcp-flags SYN,RST SYN -j TCPMSS --clamp-mss-to-pmtu
    host2:
      kind: linux
      image: alpine-networking
      exec:
        - ip link set dev eth1 mtu 1500
```

Option D: Disable TCP Offloading (When Packet Checksum Errors Occur)

In virtual environments with `veth` pairs, offload features can corrupt TCP checksums across namespaces. Disable offloading if `tcpdump -vv` shows invalid checksums:

```
docker exec host1 ethtool -K eth0 tx off rx off gso off tso off
docker exec host2 ethtool -K eth0 tx off rx off gso off tso off
```

5. Quick Troubleshooting Decision Matrix

Diagnostic Command	Expected Result (Working Path)	Action Required if Failed
<code>ping -M do -s 1472 <IP></code>	0% packet loss, RTT response	Path MTU < 1500. Lower host MTU or check tunnel/overlay overhead.
<code>ping -M do -s 9472 <IP></code>	0% packet loss (if jumbo frames enabled)	Transit link is dropping jumbo frames. Set host MTUs to 1500 or clamp MSS.
<code>tcpdump -vv -i <iface> port <p></code>	[correct] checksums on TCP data	If [bad tcp cksum], run <code>ethtool -K <iface> tx off rx off</code> .
<code>iperf3 -c <IP> -p <port></code>	Line-rate Mb/s/Gb/s throughput	If stall occurs at 0 bps, confirm end-to-end MTU uniformity and MSS clamping.